

Linear Approximation and L'Hôpital's Rule



Tangent line approximation

- 1 Use linear approximation to estimate $\sqrt{4.1}$, given $f(x) = \sqrt{x}$, $f(4) = 2$, and $f'(4) = \frac{1}{4}$.
- 2 Use linear approximation to estimate $(1.02)^5$, using $f(x) = x^5$ near $x = 1$.
- 3 Find the linearization $L(x)$ of $f(x) = \sin x$ at $x = 0$, and use it to approximate $\sin(0.1)$.

Error and over/underestimation

- 4 For $f(x) = \sqrt{x}$ (concave down for $x > 0$), explain whether the linear approximation at $x = 4$ overestimates or underestimates $f(4.1)$.
- 5 For $f(x) = x^2$ (concave up everywhere), explain whether the tangent line approximation at any point overestimates or underestimates nearby function values.

Indeterminate forms and L'Hôpital's Rule

- 6 Evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ using L'Hôpital's Rule, and confirm it matches the known limit.
- 7 Evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$ using L'Hôpital's Rule (applied twice).
- 8 Evaluate $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$ using L'Hôpital's Rule.

Recognizing when L'Hôpital's Rule applies

- 9 Explain why L'Hôpital's Rule cannot be directly applied to $\lim_{x \rightarrow 0} \frac{x+1}{x}$, and evaluate the limit by other means.

Differentials

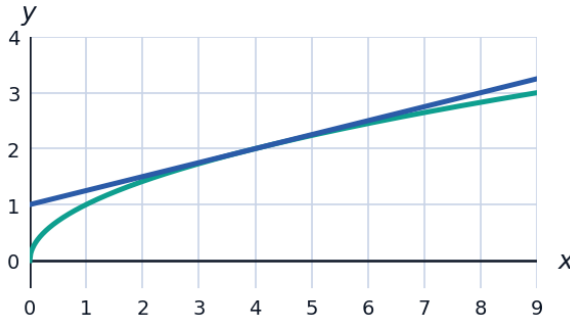
- 10 For $f(x) = x^3$, find the differential dy when $x = 2$ and $dx = 0.1$, and compare it to the actual change $\Delta y = f(2.1) - f(2)$.

More L'Hôpital's Rule practice

- 11 Evaluate $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$ using L'Hôpital's Rule (applied as many times as needed).
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Visualizing a linear approximation

- 12 The graph shows $f(x) = \sqrt{x}$ together with its tangent line at $x = 4$, $L(x) = 2 + \frac{1}{4}(x - 4)$.



L'Hôpital's Rule with exponential indeterminate forms

- 13 Evaluate $\lim_{x \rightarrow \infty} \frac{x^2}{e^x}$ using L'Hôpital's Rule (applied twice).
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Linear approximation in an applied context

- 14 The side length of a cube is measured as 5 cm, with a possible measurement error of ± 0.02 cm. Use the differential $dV = 3s^2 ds$ to estimate the maximum resulting error in the computed volume $V = s^3$.
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A capstone L'Hôpital's Rule problem

- 15 Evaluate $\lim_{x \rightarrow 0^+} x \ln x$ by first rewriting it as a quotient $\frac{\ln x}{1/x}$ to apply L'Hôpital's Rule.
- 16 A surveyor measures the angle of elevation to a tower as $\theta = 60^\circ$ with a possible error of $\pm 1^\circ$ ($\pm \frac{\pi}{180}$ radians). If the tower's height is estimated using $h = 100 \tan \theta$ (with the base 100 m away), use the differential $dh = 100 \sec^2 \theta d\theta$ to estimate the resulting error in the height, to two decimal places.
- 17 Evaluate $\lim_{x \rightarrow \infty} x \sin\left(\frac{1}{x}\right)$ by rewriting it as $\frac{\sin(1/x)}{1/x}$ and applying L'Hôpital's Rule (or recognizing the standard limit form).

- 18 For $f(x) = \cos x$ near $x = 0$, complete each step:
- a Find the linearization $L(x)$ at $x = 0$.
 - b Use $L(x)$ to estimate $\cos(0.05)$.
 - c Compare to the true value $\cos(0.05) \approx 0.99875$, and state whether the linearization over- or under-estimates.